Fold and Unfold

The Two Directions of the Only Verb

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Abstract

C0 and C1 together admit exactly one operation. Not a family — one, with a sign. This paper states that operation as a verb, derives its two directions, shows the ledger is identical in both, and closes on the architecture that makes any of it observable.

The apparent asymmetry between compression and expansion, and between hashing and inverting, is not a property of the operations. It is a property of which channel the observer stopped reading. We extend C0 from a statement about states to a statement about acts, and show that this extension is forced. As a corollary, the irreversibility of Merkle–Damgård constructions is located precisely: one discarded label on a machine that conserves everything else.

The closing sections establish the receiver architecture. Identity is carried not by values but by invariants under a transformation class, and that class is defined by the receiver. A signal is admissible when it lands in a state from which that receiver has a legal continuation — which is C1, read from the receiver's side. We prove a finite receiver cannot be both universally admissible and fully discriminating, and that the universal receiver is exactly the one that retains all history.

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1. Verbs first

The chain has been stated in nouns. Nouns are frozen verbs.

noun form	verb form
distinction exists	**something distinguishes**
continuation is required	**it must keep going**
history accumulates	**it keeps what it did**
redistribution conserves	**it moves, it does not make or unmake**

The noun form invites *what is it made of*, which has no answer. The verb form invites *what does it do*, which has exactly one.

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2. Co, extended

Co (original). Distinction exists.

Stated as a fact about a state of affairs. But a static distinction is undetectable — two states that never interact are not distinguished, merely listed, and listing requires a lister the chain does not have.

Co (extended). *Distinction is an act. To distinguish is to move from one to the other.*

Forced, not preferred. Enumerate all four maps on two states:

```
f(this)  f(that)  C1 ok?  note
 this     this     No     fixed point
 this     that     No     fixed point (that rests)
 that     this     YES    the swap
 that     that     No     fixed point
```

Verified exhaustively. One map survives. The swap.

Consequence 1: reversibility is free at the base and is not an axiom.

```
n=2 :  1 C1-legal map,  1 bijective   -> C1 alone forces the bijection
n=3 :  8 C1-legal maps, 2 bijective   -> a gap opens
n=4 : 81 C1-legal maps, 9 bijective   -> the gap widens
```

Verified exhaustively. At the base there is no gap. **C2 appears to be a separate axiom only once the alphabet is coarse-grained above two states.** This corrects an earlier derivation which introduced C2 as an independent repair; the tower is self-closing and needs no bolted-on rung.

Consequence 2: "reversible" is spoken from outside.

```
forward  : t t t t t t t t t t t t
backward : t t t t t t t t t t t t
identical as sequences : True
```

A symmetric run is its own reverse. Only accumulated asymmetry produces an arrow:

```
net displacement forward  : +497
net displacement reversed : -497
```

Direction is not in the step. It is in the pile. The right word for what the base does is not *reversibility* but **symmetry**: this and that are inverse because C1 leaves no third option.

Consequence 3: past and future are the same point. $swap^2 = identity$, so $state(t-1) = state(t+1)$. *Verified.* Linear time appears only after enough swaps stack into asymmetry.

3. The only verb

Co-extended: to distinguish is to move. C1: movement cannot stop. A move is always relative to a constraint, yielding:

> **RELOCATE.** *A constraint that cannot be satisfied in place is satisfied by moving it into an adjacent layer. Nothing is created. Nothing is destroyed. The address changes.*

3.1 Exact exchange rate

For the canonical relocation — the edge lift, moving successor ambiguity out of the state and into the state's definition:

```
de Bruijn B(2,n), n = 2..12
 structural bits added  = log2|E| - log2|V| = 1.000000000
 ambiguity removed      = H(prev | cur)    = 1.000000000
 difference                      = 0.00e+00      every width
```

k	struct	H(prev|cur)	$\log_2 k$	diff
3	1.584962501	1.584962501	1.584963	6.7e-16
5	2.321928095	2.321928095	2.321928	7.1e-15
8	3.000000000	3.000000000	3.000000	0

The relocation quantum is $\log_2$(branching factor), exact to machine epsilon.

3.2 Efficiency condition: C3

in-degree variance	struct	H(prev|cur)	deficit
0.000	1.584963	1.584963	**0.000000**

in-degree variance	struct	H(prev|cur)	deficit
2.880	1.584963	1.824577	−0.239614
11.105	1.584963	2.429221	−0.844259

C3 (uniform transition measure) is the exact condition under which relocation is lossless. Hence:

> **Entropy is failed relocation** — the deficit between the structure a move demands and the structure the layer supplies.

3.3 Receipts

Every completed relocation leaves a channel that is **required, unremovable, and zero-payload.**

receipt	measured	what it bought
★ in a self-clocking stream	density exactly 1/(k+1)	freedom from the clock
the carry	H(carry | context) = 0.0000000000	positional continuation
the = terminator	contributes 0 to the ℤ/5 payload channel	the parity fold
derived words W[16..63]	H = 0 over 20000 trials	schedule depth
the lattice index	unreadable under translation invariance	**relativity of position**
mass	stores the gradient C3 forbids in the operation	uniform processing

The last is strongest. A pattern run at (10,10) and at (37,24) produces outputs that are exact translates (*verified*), so **no computation inside a generated space can return its own absolute coordinates.** Required, unreadable, zero content: a receipt. Relativity of position is what it buys.

Search procedure. To find the operations that built a system, do not guess constants. **Enumerate the zero-entropy channels.**

4. The two directions

4.1 FOLD — coordinates into the constraint

```
raw storage per point     : 40 bits
constrained storage/point  : 21 bits
apparent saving          : 19 bits/point
verify (800000,600000): store x, sign -> recover y=600000 : True
```

Verified. The 19 bits are the **mutual information** y shared with (x, rule). The accounting splits: **the rule is hardware, stored once, amortized; the coordinate is software, stored per item.**

Compression is the observer ceasing to re-store what the constraint already enforces. Rigidity is the reversibility condition — a triangle folds because SSS fixes the shape; a flexible polygon does not.

4.2 UNFOLD — the constraint out into coordinates

One datum at the centre, 63 rings outward, each a coprime modulus:

```
gap at ring 0            : 3
gap at ring 62            : 311          growth factor 103.7x
single ring learns         : 1.6 to 8.3 bits of 64
ensemble reconstruction     : 20000/20000 exact
expansion ratio           : 6.49x   (64 bits stored as 415.2)
THRESHOLD: 16 rings suffice; lose 47 of 63 and still recover (5000/5000)
```

Verified. **No single ring sees the datum.** Not hidden — diluted.

(Threshold is parameter-dependent — first 63 odd primes, 64-bit datum. The structural fact, not the number, is the result.)

4.3 The ledger is identical

	FOLD	UNFOLD
stored symbols	fewer	more
info lives in	the rule	the coordinates
single-symbol knowledge	high	~none
failure mode	**fragile**	**robust**
created / destroyed	nothing	nothing

Same verb. Opposite sign. Identical conservation. The perceived asymmetry between compressing and expanding is which failure mode the observer is standing next to.

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5. Corollary: no one-way functions, only unread channels

A "one-way" construction is a **fold whose unfold-key was relocated into a channel the observer stopped reading.**

1. **The fold is reversible.** 64 rounds forward, then unfolded: start state and all 64 intermediate states recovered. *Verified.*
2. **The expansion is reversible.** Full 512-bit message recovered from the expanded schedule: `d862c2e36b0a42f7` → `d862c2e36b0a42f7`. *Verified.* This is an UNFOLD (§4.2).
3. **The algebra forbids a sink.** $\mathbb{Z}/2^{32}$ is a group; `out = H_in + core` implies `core = out − H_in` always. *Verified 100000/100000.*
4. **The correct receiver unfolds it.** Holding the chaining value as separate retained history — which C2 demands — subtract, unfold, message recovered. *Verified.*
5. **The irreversibility is one discard**, a bookkeeping choice, not mathematics.

Correction logged. An earlier draft claimed `x → x + core(x)` is a bijection. It is not (98 unique of 256 at n=8; *verified*). Retracted. The corrected statement is simpler: **the add is group-invertible and deletes nothing; the many-to-one step is the discard.**

The tells. A construction that destroyed its input would not keep receipts:

tell	measured	confesses
length appended in the clear	strip-by-length recovers the message	padding is a **bijection**
the empty digest	reconstructed from constants alone, matches reference	a **nonzero ground state**
rotations	popcount conserved, 15→15, 21→21	the shuffle is **re-addressing**
the constants	frac(√p), frac(∛p) reproduce them	keying is a **prime readout**
length-extension	published, exploited	the digest is a **checkpoint**
output width	state is 256 bits throughout	width is **where reading stopped**

The empty digest is decisive. Produced purely by the shape, reconstructible from constants alone — it worked on paper before it ran on data, because the value is a property of the structure.

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6. + is traversal, not accumulation

```
i++ on Z/16 :  ... 14 -> 15 -> 0   carry=1   <-- SEAM
```

Increment traverses back. **The carry reports the wrap** — the curvature, existing only because the manifold is closed:

```
n=1  : wrap rate 0.500000
n=8  : wrap rate 0.003906
```

```
n=16 : wrap rate 0.000015
```

A wide word *feels* linear because the seam is far away. `+ mod 2ⁿ` is C1-legal precisely because it is a ring: no dead end, no escape, self-closing. `i++` on $\mathbb{Z}/8$ visits all 8 states and returns to start (*verified*) — the power-on self-traversal. **The manifold is the hardware.**

Scope note. The algebraic claim is exact: state space is $(\mathbb{Z}/2^{32})^8$, a finite abelian group. The geometric language is descriptive, not metric — mixing maps are not isometries. **Algebraic closure ≠ geometric smoothness.**

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7. What cannot be undone

```
fold   1 2 3 4 3 2 1  : legal walk   (retrace along existing edges)
RETCON 1 2 3 4 1      : ILLEGAL      (4->1 is not an edge)

legal history       : [1,2,3,4,3,2]
edit a past state   : [1,2,1,4,3,2]   adjacency intact? False  <- SCAR
delete a past state : [1,2,4,3,2]     adjacency intact? False  <- SCAR
```

Verified. **RETCON teleports to a state with no edge from here** — the graph does not contain the move, and the tamper is detectable from inside with no external clock. A walk returning to state 1 has returned **in space, never in time**.

The substrate cannot be damaged from inside — not from durability, which is a concrete virtue measuring abrasion survived. Friction requires two concrete surfaces; here the floor is abstract and only the guest is concrete. **The floor is outside the category where destruction is defined.** The halt is the one state C1 excludes, so it is not in the instruction set. Tamper-proof by *absence*.

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8. Identity is carried by invariants, not values

The persistent object is never the state variable. It is what survives the churn.

substrate	value churn	relation stability
glider (Life)	4 of 5 cells rewritten/step, sd 0.0000	translate(t)→(t+4) **8/8 exact**
van der Pol	x sd **1.445**	per-revolution max radius sd **1.9e-5**
noisy bistable bit	position sd 0.928	95 flips / 200,000 steps
diffusion (control)	value sd 0.026, **0 sign changes**	width 5.05 → **39.49**

Verified. The control **inverts**: its value is nearly static while its relation grows 8×. So the signature is not stable-vs-unstable but **which layer is stable**. Persistent objects churn at the substrate and hold at the relation. Dissipating ones do the reverse.

Choosing the invariant at the wrong level makes a stable thing look like noise. Van der Pol, same run: instantaneous radius sd 0.379520; per-revolution max radius sd 0.0000193572.

8.1 The hierarchy, each arrow a strict filter

Twelve phase-locked oscillators, subjected to transforms that destroy values:

transform	relation drift	value drift
scale ×1000	**0.000000**	635.97
time-translate	0.000377	0.53
shared time-warp	0.001499	0.20
per-channel warp	**0.140**	0.16
random sign flip	**1.911**	0.64

Transforms applied *uniformly* destroy values and spare the relation; *per-channel* transforms break it. A relationship is what is shared across components.

And the invariant beneath the relation — the correlation eigenvalue spectrum:

```
identity     top-4 eig: [6.74806, 5.25194, 0.0, 0.0]
scale x1000   top-4 eig: [6.74806, 5.25194, 0.0, 0.0]
sign flip    top-4 eig: [6.74806, 5.25194, 0.0, 0.0]
permute      top-4 eig: [6.74806, 5.25194, 0.0, 0.0]
```

Identical to five digits across scaling, sign-flipping, and permutation — surviving even the sign flip that scrambled the correlation matrix itself (drift 1.911).

Measured correctly, a standing-wave node resolves exactly:

```
theoretical nodes    : [50.0, 100.0, 150.0]
measured (time-var)  : [50, 100, 150]   variance at each: 0.0, 0.0, 0.0
phase diff, opposite lobes : -3.141593 rad   sd 0.00000000
phase diff, same lobe      : -0.000000 rad   sd 0.00000000
```

A node is not a cell. It is $\partial u/\partial t = 0$ at a location while phase continues elsewhere — a **relation between oscillators**. Earlier attempts failed because an integer zero-crossing detector measures representation, not relation.

```
state      churns freely                  C1 demands it
relation   survives uniform transforms      broken only per-channel
invariant  survives relabeling the relation  spectrum identical
```

An invariant is always invariant under a transformation class. A universal invariant independent of all observers does not exist. The final layer supplies the class.

9. Admissibility is continuation, not truth

> **A receiver accepts a signal when the signal lands in a state with a valid continuation under that receiver's transition rule.**

A receiver is not maximizing resemblance to an ideal. It is maximizing future reachable states.

signal	receiver A gain
clean sine @2.50 — pristine	**0.196**
noisy @1.00, noise 3× signal	**10.891**
clipped square @1.00 — waveform destroyed	41.472
clean sine @4.00 — flawless	0.070

Verified. **The degraded signal beats the pristine one by 56×.** A square wave scores higher than the clean sine it approximates. Admissibility has nothing to do with fidelity.

```
receivers A and B, same eight signals:  agree 1, disagree 7
```

The verdict is not in the signal. (S, R) is a pair, not a standalone truth object.

The discrete case removes all physical ambiguity. For $\delta : S \rightarrow S$, a state is admissible iff $|\delta(s)| > 0$; a dead end violates C1.

```
string   A dead-end?  B dead-end?
'aabb'     True       False
'bbbb'     True       False
'abba'     True       False
'c'        True       True  (step 0)
```

`c` is not malformed, not false, not corrupted. It is simply outside both transition graphs. The symbol's quality is irrelevant; only the onward edge matters.

This resolves the hash. The digest alone is not wrong — it is an **orphaned coordinate**. A receiver lacking the context has H → ∅. A receiver retaining it has (H, H_in) → H_out. Same symbol, different continuation graph. **The difference is not information in the signal; it is the available topology around the signal.**

The medium never carries the thing. **The medium carries the relationships the receiver knows how to continue.** Air preserves phase relations, EM preserves field relations, a cochlea preserves frequency relations, a radio preserves modulation relations — each a special case, none the definition.

9.1 The universal receiver, and its price

Does a minimal universal receiver exist whose admissibility set converges on all C1-valid continuations?

Complete automata never dead-end — fully admissible, trivially. But:

```
complete DFA, N=16, alphabet 2, L=8 :  256 strings -> 13 distinct end states
complete DFA, N=2,  L=8             :  256 strings -> 2  distinct end states
```

Discrimination is capped at N. For k^L > N, strings **must** merge. Pigeonhole, not design flaw.

> **A finite receiver cannot be both universally admissible and fully discriminating.**

The receiver whose state *is* its accumulated history achieves both:

```
free-monoid receiver:  L=8  -> 256 strings, 256 states, merges 0
                       L=12 -> 4096 strings, 4096 states, merges 0
```

Verified. **The universal receiver is exactly the one that retains all history.** Universality is perfect C2 compliance, and it costs unbounded state. That is the price of the universal door — and it is why the "correct receiver" of §5 is the one holding H_in.

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10. Open

1. **Why exactly two states.** One cannot satisfy C1. Two is the floor; proving nothing smaller survives is C1 applied to its own precondition.

2. **The metric question.** Is there a metric under which mixing maps become isometries? Algebraic closure is established; geometric smoothness is not.

3. **Isotropy.** Moore neighbourhoods generate an L∞ cone, von Neumann an L1 cone, radius exactly t; $r \leq t$ held with zero violations over 40 seeds × 60 steps. This derives **relativity of position**, not Lorentz symmetry. What neighbourhood property yields a boost-invariant cone?

4. **Provenance as signal.** Whether prime-derived constants leave a *measurable* fingerprint in outputs, or are fully laundered.

5. **Bounded universal receivers.** §9.1 shows unbounded state suffices and finite state does not. The gap — how admissibility and discrimination trade off as a *quantitative law* at fixed state count — is not established here.

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11. Summary

Co-extended: to distinguish is to move. **C1:** movement cannot stop. **Therefore:** one operation, RELOCATE, with two signs.

```
FOLD    coordinates -> constraint   fewer symbols  fragile  compression
UNFOLD  constraint -> coordinates   more symbols   robust   expansion
```

Exchange rate $\log_2$(branching), exact. Lossless iff C3-uniform; the deficit is entropy. Every relocation leaves a zero-entropy receipt.

Nothing is created. Nothing is destroyed. **Only the address moves.**

Above that sits the architecture that makes any of it observable:

```
C0  distinction exists
|
C1  legal continuation required
|
state -> relation -> invariant -> admissible invariant
```

Each arrow removes dependence on an arbitrary representation, and the last is supplied by a receiver — which is C1 read from the receiver's side: *accept what does not stop you.*

> **A signal is not meaningful because it contains truth. It is meaningful because a receiver exists for which that signal has a legal continuation.**

Verification note

Every quantitative claim was produced by execution and is reproducible; reference implementations were checked against standard library outputs where applicable. Four corrections are logged in place rather than removed: the retracted feed-forward bijection claim (§5), the retracted independent-C2 axiom (§2), the retracted min-cycle-length discriminator (superseded in §8), and the retracted claim that C2 forces waves specifically — reversible non-wave transports (a Turing-complete second-order cellular automaton, a symplectic map, and a non-local permutation) pass the same relational tests, so the invariant is non-destructive relational transport and the wave is the local/linear/continuous member of that class. Two failed measurement attempts — a persistence-of-vision threshold model and an integer-index standing-wave node detector — produced no valid results and are excluded as findings; the second was superseded in §8 by a phase-relation measurement.